

Generator#

<https://pytorch.org/docs/stable/generated/torch.Generator.html>

Description:

Creates and returns a generator object that manages the state of the algorithm which produces pseudo random numbers. Used as a keyword argument in many In-place random sampling functions. device (torch.device, optional) – the desired device for the generator. An torch.Generator object.

Function Signature

```
class torch.Generator(device='cpu')#
```

Parameters

Returns

Return type

```
clone_state() → torch.Generator#
```

Returns:

Generator

Code Examples

```
>>> g_cpu = torch.Generator()
>>> g_cuda = torch.Generator(device='cuda')
```

```
>>> g_cuda = torch.Generator(device='cuda')
>>> cloned_state = g_cuda.clone_state()
```

```
>>> g_cpu = torch.Generator()
>>> g_cpu.device
device(type='cpu')
```

```
>>> g_cpu = torch.Generator()
>>> g_cpu.get_state()
```

```
>>> g_cuda = torch.Generator(device='cuda')
>>> current_state = g_cuda.graphsafe_get_state()
```

```
>>> g_cuda = torch.Generator(device='cuda')
>>> g_cuda_other = torch.Generator(device='cuda')
>>> current_state = g_cuda_other.graphsafe_get_state()
>>> g_cuda.graphsafe_set_state(current_state)
```

```
>>> g_cpu = torch.Generator()
>>> g_cpu.initial_seed()
2147483647
```

```
>>> g_cpu = torch.Generator()
>>> g_cpu.manual_seed(2147483647)
```

```
>>> g_cpu = torch.Generator()
>>> g_cpu.seed()
1516516984916
```

```
>>> g_cpu = torch.Generator()
>>> g_cpu_other = torch.Generator()
>>> g_cpu.set_state(g_cpu_other.get_state())
```

Detailed Documentation

Documentation

Creates and returns a generator object that manages the state of the algorithm which produces pseudo random numbers. Used as a keyword argument in many In-place random sampling functions.

device (torch.device, optional) – the desired device for the generator.

An torch.Generator object.

Clones the current state of the generator and returns a new generator pointing to this cloned state. This method is beneficial for preserving a particular state of a generator to restore at a later point.

A Generator pointing to the newly cloned state.

Generator.device -> device

Gets the current device of the generator.

Returns the Generator state as a torch.ByteTensor.

A torch.ByteTensor which contains all the necessary bits to restore a Generator to a specific point in time.

Retrieves the current state of the generator in a manner that is safe for graph capture. This method is crucial for ensuring that the generator's state can be captured in the CUDA graph.

A Generator point to the current state of the generator

Sets the state of the generator to the specified state in a manner that is safe for use in graph capture. This method is crucial for ensuring that the generator's state can be captured in the CUDA graph.

state (torch.Generator) – A Generator point to the new state for the generator, typically obtained from `graphsafe_get_state`.

Returns the initial seed for generating random numbers.

Sets the seed for generating random numbers. Returns a torch.Generator object. Any 32-bit integer is a valid seed.

seed (int) – The desired seed. Value must be within the inclusive range `[-0x8000_0000_0000_0000, 0xffff_ffff_ffff_ffff]`. Otherwise, a `RuntimeError` is raised. Negative inputs are remapped to positive values with the formula `0xffff_ffff_ffff_ffff + seed`.

An torch.Generator object.

Gets a non-deterministic random number from `std::random_device` or the current time and uses it to seed a Generator.

Sets the Generator state.

new_state (torch.ByteTensor) – The desired state.